

ENTERPRISE BANKING INTELLIGENCE & EARLY WARNING PLATFORM

(EBIEWP)

*End-to-End Business Intelligence, Customer 360,
Early Warning & Operational Automation*

Core principle: Detect → Understand → Prioritize → Route → Act

Project Documentation

Business & Technical Case Study

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1. Executive Summary

The Enterprise Banking Intelligence & Early Warning Platform (EBIEWP) is an end-to-end banking Business Intelligence and decision-intelligence solution designed to demonstrate how operational banking data can be transformed into trusted analytical insight and connected directly to operational intervention.

The platform combines synthetic data generation, SQL Server data warehousing, ETL, dimensional modelling, analytical SQL, Power BI, DAX, Customer 360 intelligence, early-warning classification, risk prioritization, stakeholder routing and workflow automation.

2. Business Problem

Customer deterioration can manifest through multiple signals across banking operations. Credit behaviour, repayment performance and customer service experience may each provide partial evidence of deterioration.

When these signals are analyzed independently, decision-makers may struggle to determine which customers require attention, why they are being flagged, how severe the situation is, what action should be taken, who is responsible and how the stakeholder should be notified.

- Increasing Days Past Due
- Consecutive missed payments
- Deteriorating payment behaviour
- Increasing complaint frequency
- Unsatisfied complaints
- Repeat complaints
- Service deterioration
- Credit deterioration
- Declining customer health
- Increasing financial exposure

The platform addresses these questions:

- Which customers require attention?
- Why are they being flagged?
- How severe is the situation?
- What action should be taken?
- Which stakeholder is responsible?
- How can the stakeholder be notified quickly?

3. Business Objectives

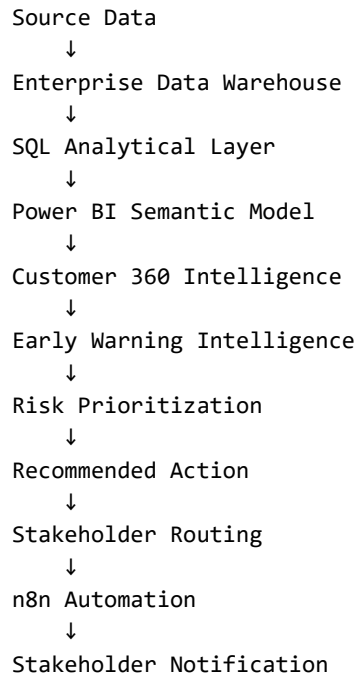
EBIEWP was designed to build trusted enterprise BI while connecting analytical findings to operational intervention.

- Build an enterprise-grade dimensional data warehouse.
- Centralize banking data across multiple business domains.
- Deliver trusted data for executive decision-making.
- Monitor customer and lending performance.

- Detect emerging customer risks.
- Prioritize customers requiring intervention.
- Recommend appropriate actions.
- Route alerts to responsible stakeholders.
- Automate stakeholder notification.

4. Solution Architecture

The solution follows a layered Business Intelligence architecture that separates data engineering, analytical modelling, visualization and operational workflow concerns.



5. Data Architecture & Warehouse

The SQL Server layer provides the enterprise data warehouse foundation. The warehouse uses dimensional modelling with customer, account, branch and date dimensions supported by transaction, loan, repayment and complaint facts.

Dimension	Purpose
DimCustomer	Customer demographics, segmentation and banking profile
DimAccount	Customer account portfolio
DimBranch	Branch hierarchy and geographic information
DimDate	Enterprise calendar dimension

Fact	Purpose
FactTransaction	Banking transaction history
FactLoan	Lending portfolio
FactRepayment	Loan repayment monitoring
FactComplaint	Customer experience and complaint lifecycle

6. Data Engineering & Validation

Synthetic banking datasets are generated programmatically and transformed through a structured SQL Server ETL process. Validation is applied before downstream reporting.

- Dataset validation
- Row count validation
- Duplicate detection
- Mandatory field validation
- Referential integrity validation
- Banking business-rule validation
- Semantic layer validation
- Stored procedure validation
- Warehouse health validation

7. SQL Analytical Layer

The SQL implementation provides the analytical foundation for downstream BI through warehouse tables, analytical views, stored procedures, keys, indexes and business rules.

- Database creation
- Dimension and fact tables
- Data loading and transformations
- Primary and foreign keys
- Performance indexes
- Analytical SQL views
- Stored procedures
- Business-rule enforcement

8. Power BI Semantic Model

The Power BI semantic model provides the business-facing analytical layer, combining warehouse data with DAX measures and calculated logic.

- Customer analytics
- Account and deposit analysis
- Loan and credit analysis
- Repayment behaviour
- Complaint intelligence
- Branch performance
- Customer health
- Early Warning Intelligence
- Recommended intervention

- Stakeholder routing

9. Power BI Dashboard

The Power BI report is a multi-page executive and operational analytics application covering executive banking performance, Customer 360, lending, credit risk, branch performance, complaint intelligence and Early Warning Intelligence.

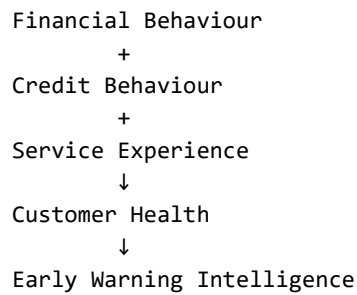
Dashboard Area	Purpose
Executive Overview	High-level banking performance and customer/business health
Customer Intelligence / Customer 360	Unified customer relationship and behavioural analysis
Revenue & Profitability	Overall bank revenue and profitability status
Credit & Loan Risk	Repayment deterioration, DPD, missed payments and exposure; Lending exposure, products, status and portfolio performance
Branch & Operational Performance	Branch-level customer and financial performance
Early Warning Intelligence	Customers requiring attention, warning drivers and recommended action

- Slicers for relevant business dimensions
- Cross-visual filtering where analytically appropriate
- Page navigation controls
- Customer-level investigation
- Operational filtering
- Consistent purple visual theme

10. Customer 360 Intelligence

Customer 360 consolidates customer-level information across financial, credit and service domains to provide a holistic view of the customer relationship.

- Customer identity and profile
- Branch and region
- Customer segment
- Account relationships
- Deposits and transaction activity
- Loan exposure and outstanding principal
- Payment behaviour
- Days Past Due
- Consecutive missed payments
- Complaint volume
- Unsatisfied complaints
- Repeat complaints
- Customer health



11. Early Warning Intelligence

The Early Warning Intelligence layer converts customer financial and service signals into structured, explainable warnings that can be prioritized and routed to the appropriate stakeholder.

Customer Health

- Healthy
- Watchlist
- At Risk

Warning Type

- Credit Deterioration
- Service Deterioration
- Credit + Service

Warning Severity

- High
- Medium

Key Signals

- Days Past Due
- Consecutive missed payments
- Payment behaviour
- Outstanding principal
- Total complaints
- Repeat complaints
- Unsatisfied complaints

The Early Warning Automation table provides the operational dataset for the automation workflow, including customer, branch, health, repayment, complaint, warning, action, exposure and stakeholder-routing fields.

12. Business Logic & Risk Prioritization

The rules-based intelligence framework combines multiple signals to support explainable decisions and prioritizes customers requiring attention.

Priority Ranking

- Warning severity

- Days Past Due
- Complaint volume
- Outstanding principal

Recommended Intervention

- Relationship Manager Follow-up
- Customer Service Follow-up
- Service Recovery

13. Stakeholder Routing

Once a customer is classified for intervention, the platform identifies the responsible stakeholder and associated email destination.

Recommended Action	Stakeholder
Relationship Manager Follow-up	Branch Manager
Customer Service Follow-up	Customer Service
Service Recovery	Service Recovery Team

14. Automation Platform Selection: n8n vs. Power Automate

Power Automate was considered because EBIWEP is built around Microsoft technologies. However, the automation demonstration was implemented using n8n because the project was developed in a personal environment rather than an organizational Microsoft work or school tenant.

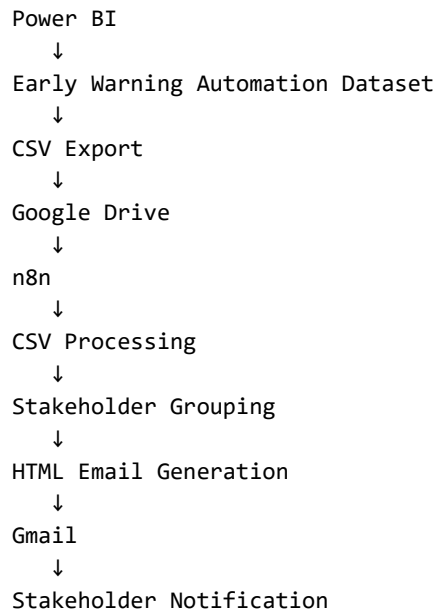
n8n provided a practical way to demonstrate the BI-to-action workflow without making the demonstration dependent on enterprise Microsoft account, tenant and licensing arrangements. The decision does not imply that n8n is inherently superior to Power Automate. In a production organization already standardized on Microsoft 365, Power Automate would be a strong alternative because of its native Microsoft ecosystem integration.

Consideration	n8n	Power Automate
Personal development environment	Strong fit	More dependent on organizational Microsoft environment
Workflow flexibility	High	High
Microsoft ecosystem integration	Good	Excellent
Email automation	Supported	Supported
Self-hosting option	Yes	Primarily Microsoft-managed service
EBIEWP demonstration	Selected	Considered alternative

Decision: n8n was selected because it allowed the automation concept to be demonstrated effectively within the available personal development environment while keeping the workflow architecture transferable.

15. n8n Automation Workflow

The automation layer connects the Early Warning Automation dataset to stakeholder notification using n8n, Google Drive and Gmail.



- Schedule trigger
- Retrieve the exported dataset from Google Drive
- Process CSV records
- Group alerts by stakeholder
- Generate stakeholder-specific HTML email
- Send through Gmail

16. Automated Stakeholder Email

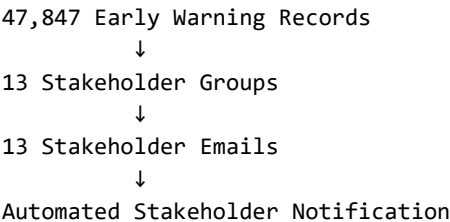
The email-generation logic produces a stakeholder-specific HTML notification containing alert summaries, financial exposure, warning drivers, recommended action and prioritized customer cases.

- Stakeholder identification
- Total alert count
- High-severity count
- Medium-severity count
- At-risk count
- Watchlist count
- Total outstanding principal
- Leading warning drivers
- Recommended action
- Prioritized customer table

17. Automation Demonstration & Results

The final automation demonstration used the Early Warning Automation dataset and successfully produced stakeholder-specific notifications.

Metric	Result
Early Warning Records	47,847
Stakeholder Routing Groups	13
Stakeholder Emails Generated	13
Stakeholder Emails Delivered	13



18. Project Scale & Statistics

The project was built at a scale intended to demonstrate enterprise BI concepts and workflow automation.

Component	Volume
Customers	100,000
Accounts	153,035
Branches	150
Calendar Dates	3,834
Transactions	100M+
Active Loans	4,937
Repayment Records	33,000+
Complaint Records	1.13M+
Early Warning Records	47,847

19. Limitations

EBIEWP is a portfolio demonstration rather than a production banking implementation. The underlying data is synthetic and production controls are outside the scope of the demonstration.

- Synthetic data rather than live banking data
- Demonstration automation infrastructure
- No enterprise identity and access management
- No production secrets management
- No real-time streaming ingestion
- No persistent alert deduplication
- No production case-management integration
- No production SLA monitoring
- No production audit infrastructure

20. Production Considerations

A production banking implementation would require additional controls around security, governance, monitoring, resilience and operational case management.

- Enterprise identity and role-based access control
- Secure secrets management
- Persistent alert state
- Alert acknowledgement and resolution tracking
- Audit logging
- Data lineage
- Pipeline monitoring
- Automation failure monitoring
- SLA monitoring
- Enterprise case-management integration
- Model governance

21. Future Enhancements

Potential enhancements include persistent alert state, deduplication, intervention tracking, real-time ingestion, advanced risk scoring and enterprise monitoring.

- Real-time data ingestion
- Persistent alert deduplication
- Intervention history
- Stakeholder acknowledgement
- CRM/case-management integration
- Advanced statistical or machine-learning risk scoring
- Automated escalation
- Enterprise monitoring and observability

22. Final Project Outcome

EBIEWP demonstrates a transition from traditional descriptive Business Intelligence to operational decision intelligence.

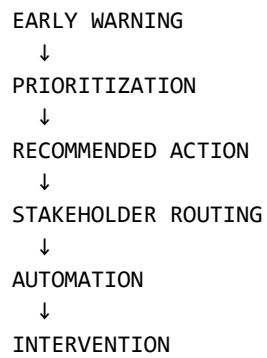
- What is happening?
- Which customers require attention?
- Why are they being flagged?
- How serious is the situation?
- What should happen next?
- Who should act?
- How can the stakeholder be notified automatically?

DATA

↓

CUSTOMER 360

↓



23. Conclusion

The platform connects enterprise data engineering, Customer 360 analytics, early-warning intelligence, risk prioritization, stakeholder routing and workflow automation into a coherent end-to-end business solution.

Detect → Understand → Prioritize → Route → Act

Appendix A — Technology Stack

Layer	Technology
Programming	Python
Data Processing	Pandas
Synthetic Data	Faker
Database	SQL Server 2022, 2026
ETL	SQL
Data Warehouse	Star Schema
Analytics	SQL / DAX
Business Intelligence	Power BI
Automation	n8n
File Storage	Google Drive
Email Delivery	Gmail
Version Control	Git / GitHub

Appendix B — Author

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